Uriel Federez

Full-stack Software Engineer · Python/Django, React/TypeScript, React Native · AI Integration · Philippines, UTC+8

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Skills

Languages	Python, TypeScript, JavaScript, HTML, CSS, SQL, C++
Backend	Django, Django REST Framework, Django Channels, FastAPI, Celery, Redis, PostgreSQL, PgBouncer
Frontend & Mobile	React, React Native, Vite, Tailwind CSS, Storybook
Infrastructure	Docker, Azure, Terraform, GitLab CI/CD, DigitalOcean, Prometheus, Grafana
AI Services	Azure Document Intelligence, Unstructured, Speechmatics, MLKit
Graphics	OpenGL, GLSL
Testing	Cypress, pytest

Experience

Mahusai *Remote*

Full-stack Software Engineer *Jan 2023 - Present*

- **Audiobook QA Platform** (media client) · Django, React, TypeScript, Celery, Azure AI
  - Co-built an AI-assisted QA platform that flags manuscript/audiobook mismatches; adopted across the client's entire QA workflow, with ~90 daily users processing 88 books (~880 hours of audio) last month.
  - Developed logic to reduce false positives using pronunciation assessment and fuzzy text matching heuristics, filtering out 320K+ flagged discrepancies across 261 projects, and reducing the number of items requiring manual review by 64%.
  - Built the React interface for project tracking, client visibility, report generation, and proofing, with audio playback that highlights each spoken word and detected discrepancy in real time.
  - Built a FastAPI microservice that exports discrepancies as MIDI markers to use with existing audio editing software.
  - Built real-time WebSocket chat and per-project discussion boards (Django Channels, Redis) with rich-text messages (Tiptap / ProseMirror JSON, validated server-side) and inline audio playback for pasted audio links.
  - Eliminated recurring connection failures by diagnosing gunicorn/Postgres connection exhaustion via Azure logs; tuned gunicorn concurrency and prepared a PgBouncer rollout via Terraform.
- **Dental Cosmetic Preview App** · Proof of Concept (healthcare client) · React Native, TypeScript, Django
  - Co-built a React Native PoC (camera capture, face-guided positioning, image comparison, practice management), reaching full parity in ~10 weeks with native iOS/Android apps built over the prior year.
  - Mapped face-detection coordinates to screen space, accounting for sensor rotation, platform orientation conventions, mirroring, cropping, and pixel density on Android and iOS.
  - Improved UX by adding positioning tolerance and auto-capture countdown, removing flickering warnings and manual capture.
  - Built Django endpoints sending patient images to an external AI service for treatment-outcome previews, including EXIF orientation correction.
- **Internal Company Portal** · Django REST Framework, React, TypeScript, GitLab CI, Celery, Redis
  - Built version 1 of the company's internal portal for leave requests, project allocation, and notifications, adopted for company-wide use.
  - Set up GitLab CI pipelines running automated frontend, backend, and Cypress end-to-end tests.

Projects

CPU Raytracer (C++)

Personal Project | https://github.com/urielfederez/Raytracer

- Multithreaded path tracer with live render progress preview, HDR image-based lighting, triangle meshes, and diffuse/metallic/refractive/emissive materials.

Self-hosted infrastructure

Personal Project

- Forgejo and Headscale on DigitalOcean, private Prometheus/Grafana monitoring over a tailnet, nightly encrypted restic backups to S3.

Education

Mapúa University *Makati, Metro Manila*

BS Computer Science *Aug 2019 - Sep 2022*

- Graduated *magna cum laude*